

CamARa: Exploring and Creating Camera Movements with Spatial Reference in Augmented Reality

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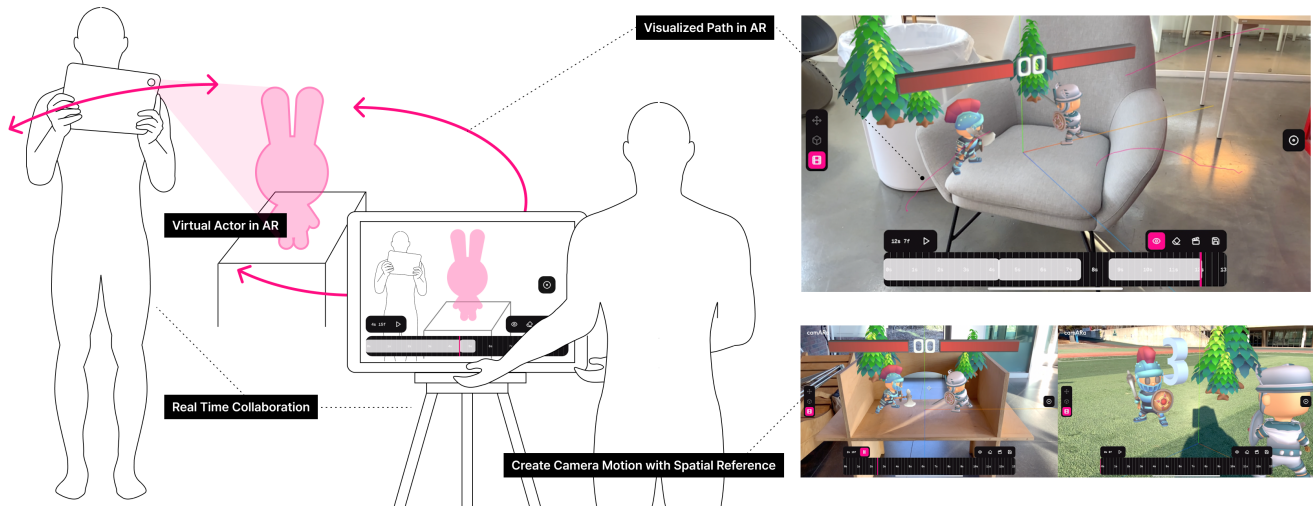


Figure 1: CamARa is an augmented camera layout system for exploring and creating camera motion utilizing spatial reference in Augmented Reality.

Abstract

Camera layout is a critical aspect of digital production, shaping the narrative and emotional tone of animation, CGI, and previsualization. However, managing camera movements in 3D space is a known challenge due to the complexity of controlling six degrees of freedom (DoF). Virtual camera systems aim to replicate the functionality of physical cameras, allowing users to manipulate a virtual camera within a digital scene. However, these systems often struggle with environmental constraints and can lead to spatial disorientation. To address these limitations, we present CamARa, an augmented camera layout system that leverages augmented reality (AR) for intuitive, spatially aware interaction. CamARa allows users to explore and create camera movements with a mobile device, utilizing spatial reference in the environment. It visualizes

movement trajectories in AR, facilitating collaboration and iterative design. This work highlights the potential of AR as a versatile alternative to traditional and VR-based camera layout tools, bridging the gap between physical and virtual spaces to support digital cinematographic workflows.

CCS Concepts

• **Human-centered computing** → **User interface programming; Mixed / augmented reality.**

Keywords

camera layout, augmented reality, virtual camera, spatial reference

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04
<https://doi.org/10.1145/3706599.3721180>

ACM Reference Format:

Hyewon Lee, Christopher Bannon, and Andrea Bianchi. 2025. CamARa: Exploring and Creating Camera Movements with Spatial Reference in Augmented Reality. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3706599.3721180>

1 Introduction

Camera layout is a crucial phase in cinematography, as it determines the message and emotion that a scene aims to convey in both live-action films and animation [4]. For any given scene, the style and mood can change significantly based on the numerous combinations of different shots. In digital production for animation, CGI, and previsualization, managing the camera within a 3D scene presents a complex challenge due to its numerous Degrees of Freedom (DOF): pitch, yaw, roll, surge, heave, and sway [2]. This complexity can make it difficult to effectively express ideas, requiring laborious and repetitive tasks, regardless of proficiency. To tackle this issue, prior studies have investigated various control metaphors to overcome the challenges of operating a 6DoF camera with 2D devices such as a mouse [14].

One practical approach, based on the idea of replicating real-world camera functionality, is a virtual camera system [1, 13]. This system allows users to generate camera movements through familiar interactions, such as physically moving a handheld device (HHD) as if operating a real camera [9]. The device's motion is tracked and applied to a virtual camera, with the resulting view rendered on the device's screen. While this approach enables intuitive control, it does not incorporate physical space as context, creating a disconnect between the physical and virtual. Users must navigate the real world while focusing on the virtual view displayed on the screen, which can lead to spatial disorientation and disrupt their ability to account for the constraints and layout of the surrounding space. Furthermore, without physical references, it becomes challenging to trace the paths of previous cameras or other cameras and understand how they were generated, complicating collaboration.

To empower designers with contextual awareness of their physical surroundings and to utilize these as references for camera motion, we propose CamARA, an augmented virtual camera layout system for exploring and creating camera movements in Augmented Reality (AR). It includes importing a 3D scene into an AR environment, creating desired camera movements, and exporting the camera with motion back to the original 3D scene. CamARA enables designers to utilize spatial references from the real world through an AR scene. Unlike existing virtual camera systems, CamARA empowers designers to integrate their physical surroundings into the creative workflow, turning typical physical constraints into context. Designers can set up 3D scenes using real objects as reference points and tailor the environment to the scene's motion, creating an intuitive and immersive experience. Sharing the same physical references enables real-time collaboration and CamARA visualizes camera paths in AR, allowing designers to build upon existing motions and foster creative exploration. This system is built as a web-based interface for mobile devices and seamlessly integrates with external 3D animation tools, enabling a smooth transition from AR to digital production workflows.

2 Related Work

2.1 Automated Camera Layout System

Prior research has explored methods to simplify the complexity of camera layouts through automated systems [5]. These systems automatically generate camera movements for a given 3D scene. Previous studies have examined various computational techniques

to create camera motions that consider the 3D environment, including character animation [10, 12]. To achieve a polished camera layout, many systems learn from reference videos using machine learning techniques that follow traditional cinematography principles [6, 7, 16]. However, automated systems can limit individuals' ability to explore camera layouts, restricting their opportunity to express style, creativity, and collaboration. Additionally, these approaches often require expensive computing resources. Our work aims to provide accessible control through personal devices, allowing users the freedom to experiment and express their ideas.

2.2 AR/VR in Previsualization Tool

Previsualization (previs) is a collaborative process that creates preliminary versions of shots, typically in 3D [15]. Through previs, filmmakers visually explore and communicate creative ideas while planning technical solutions, including layout, animation, lighting, and camera work. To provide an immersive and effective previs experience, researchers leverage AR and VR technologies to control the camera and create camera motion [3, 11, 17]. Immersive VR tools that use head-mounted displays (HMDs) are often less preferred due to their low accessibility and high complexity [8]. Some previous studies have employed handheld devices (HHDs) as proxies for virtual cameras in 3D scenes, capturing the motion of these devices [1, 8, 17]. Unlike prior VR works [1, 9, 13], CamARA leverages the physical context of the real world to create camera motion on top of a virtual camera, enabling users to consider the constraints and possibilities of moving within the surrounding space. Unlike existing AR designs [17], we focus on the interaction in crafting camera motion, fostering collaboration and exploration through path visualization.

3 System Design

3.1 Interface

CamARA consists of two screens: Home and Session. On the **Home Screen (A, Figure 2)**, users can check the compatibility of their device and browser for the AR session, establish a socket connection with a 3D animation tool, and receive a 3D scene from the tool. The **Session Screen (B, Figure 2)** allows users to switch between three modes: *Hit mode*, *Anchor mode*, and *Record mode*.

In **Hit mode (B-1, Figure 2)**, users can detect the floor and place the 3D scene in the real world with the help of a guide circle. **Anchor mode (B-2, Figure 2)** allows users to adjust the orientation, location, and scale of the scene using drag and pinch gestures.

Record mode (B-3, Figure 2) displays a control interface along with a timeline bar, allowing users to play, pause, and interact with the animation. When the user presses the record button, the animation starts plays from the current frame, and the device's movements are captured. These movements are represented on the timeline bar and visually indicated in the AR environment as a path line. Users can also preview the recorded camera motions.

3.2 Walk-through: Bunny in the Forest

To demonstrate how designers can use physical references for exploring and creating camera motion in AR, we describe a user scenario with two animators, Hana and Yuna, who are using CamARA on their tablets.

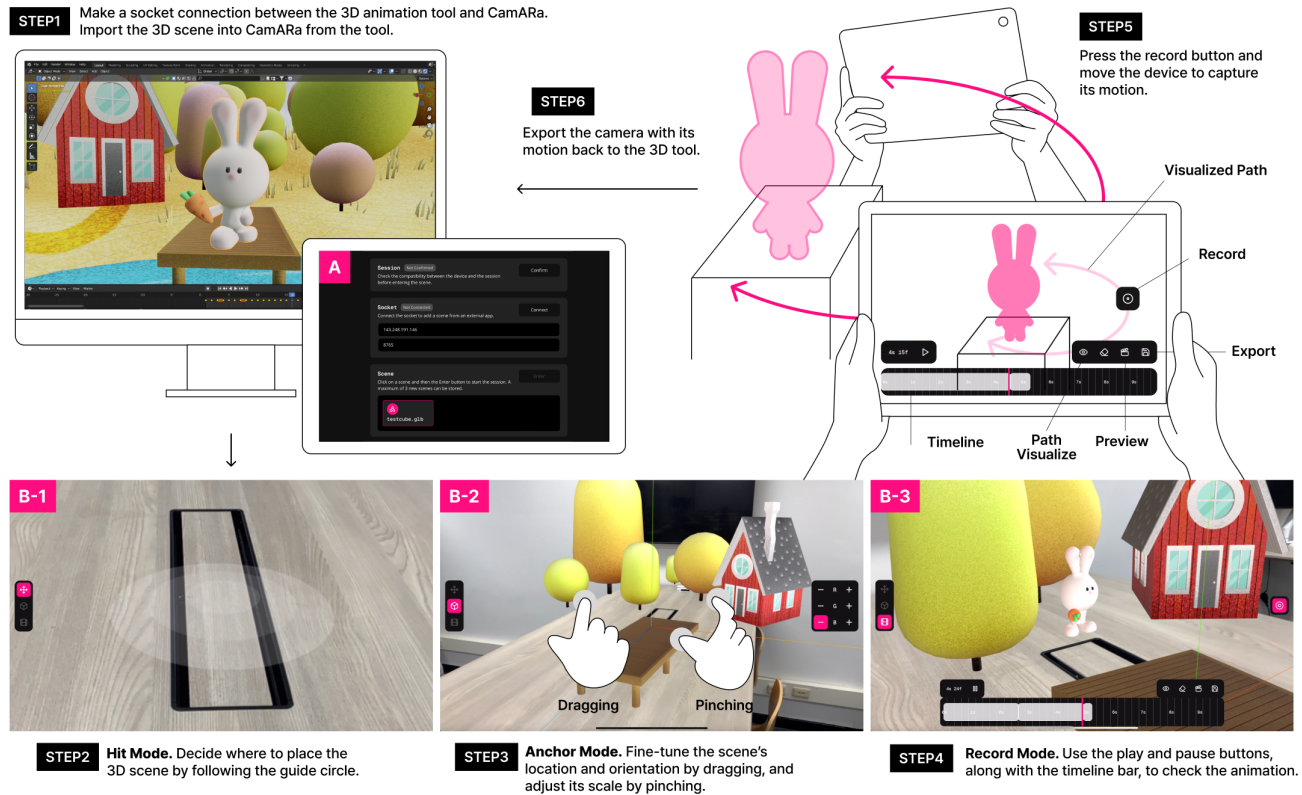


Figure 2: User flow of CamARA (a) Home screen (b) Session screen: (b-1) Hit mode (b-2) Anchor mode (b-3) Record mode

Hana and Yuna are working on the animation *Bunny in the Forest*. After they finish the character animation, they open CamARA on their tablets and connect to the 3D animation tool. After each device receives the 3D scene from the tool, they enter AR mode.

Bunny in the Forest: One sunny day, a bunny was walking in the forest and saw a carrot floating on a lake. As it reached out to grab the carrot, the bunny fell into the water.

3.2.1 Explore Shot Combinations. Before camera composition, Hana and Yuna discuss what kinds of shots they will capture. They want to visualize each idea to assess its expressiveness and narrative delivery. First, they set up their 3D scene on a table with enough space to freely move around. They adjust the anchor, ensuring that the orientation and scale of the scene to fit the space appropriately. Next, they each create camera paths by repeatedly recording and pausing within the animation's timeline, resulting in several shots captured by one camera (A, Figure 3). They iterate multiple times, previewing and refining each other's ideas as needed, and ultimately decide to go with Yuna's concept (B, Figure 3).

3.2.2 Take Detailed Shots Referencing the Guide Path. Hana and Yuna are each responsible for creating two shots. Hana uses a tripod, while Yuna holds a tablet to create camera paths for each shot. Yuna extracts the initial guide path she created, which Hana then imports. Using the visualized guide path, Hana can easily understand how Yuna moved around the table and captured each

shot. They navigate to the guide points visible in AR and develop their shots by moving relative to the table, utilizing surrounding objects like chairs to enhance their composition (A, Figure 4).

3.2.3 Take Dynamic Shots Referencing Physical Space. Hana decides to create a more appealing shot of the rabbit running by filming from the rabbit's point of view. However, this was challenging to achieve on the table, and the room was too small for the desired effect. Hana moves to the hallway, repositions the 3D scene, and aligns the animation so the rabbit runs along the floor of the hallway. Hana begins recording his movements facing the carrot, simulating the rabbit's perspective (B, Figure 4). Once Hana and Yuna are satisfied with all the shots, they export the cameras with their motion and proceed to editing and post-processing in the 3D animation tool.

4 Implementation

4.1 Technical Details

CamARA is a progressive web application (PWA) designed for AR-based camera layout and motion creation. The system's core is implemented using Three.js and TypeScript, with WebXR API integration for AR functionality. WebXR requires WebXR Viewer for iOS devices and Chrome for Android. The system facilitates seamless integration with standard 3D modeling software such as Blender and Maya by supporting GLB file workflows, a web standard for 3D objects. Users can import the 3D scenes as GLB

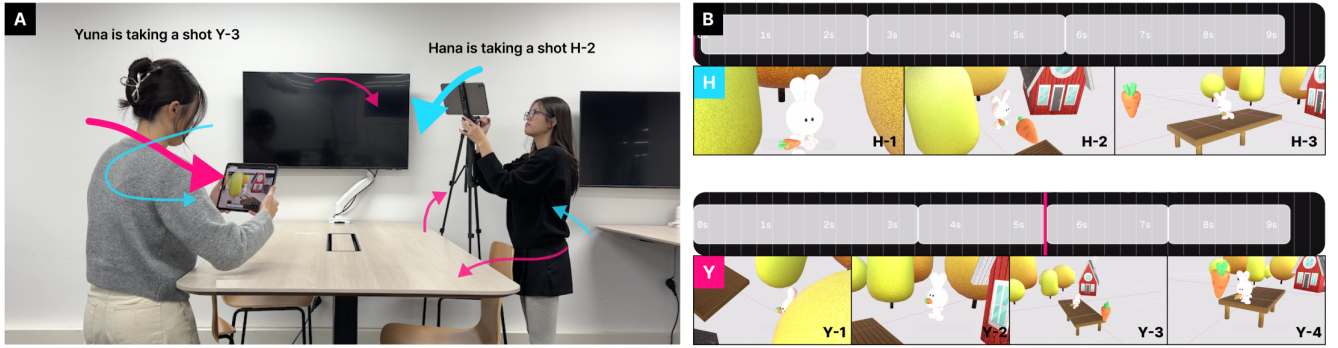


Figure 3: Explore shot combinations with CamARa (a) Two designers capture multiple rough shots in a single camera path to evaluate the coherence, expressiveness, and impact of the shot combination. (b) They preview the scene to assess its overall mood and emotion.

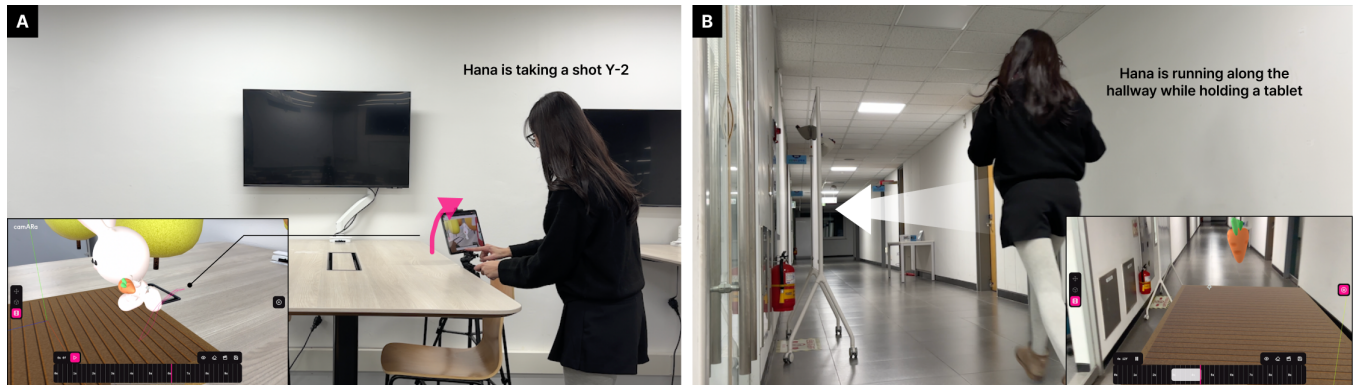


Figure 4: Develop polished shots with CamARa using spatial reference (a) A designer films a shot based on a shared guide path that others have visualized in AR. (b) A designer utilizes the spatial structure to capture dynamic shots.

files, add camera data within the AR environment, and export the enhanced files back into their preferred 3D tools. Additionally, CamARa features live server connectivity through the WebSocket protocol, enabling real-time interaction with external software and improving collaborative workflows.

4.2 Demo: Bunny in the Forest

In this interactivity, we will showcase our walk-through scenario, *Bunny in the Forest*, on an iPad running CamARa through WebXR Viewer. To enable live server connectivity, we developed CamARa and Blender¹, a free software for 3D modeling and animation.

Participants begin by importing the 3D scene from Blender and then proceed to exploring and creating camera motion in AR. Participants will be instructed on the hit, anchor, and record modes, then left to experiment on their own to create camera compositions. The demonstration area provides space for physical movement, and participants will be encouraged to walk around the space to take various shots from different angles. Additionally, emphasis will be placed on exploring dynamic camera motions, such as shaking, jumping, or thrusting to create unique camera paths. If they want to

take sophisticated shots they are allowed to use a provided tripod. Participants can take as many recordings as they like and preview their recorded work in the app and view their recorded camera path in AR. A monitor will be set up for participants to view their camera path in blender if they choose to export. Finally, the participant's screen will be shared on a monitor in real time to allow spectators to participate in the interactivity session as well.

5 Conclusion

We present CamARa, an augmented virtual camera system for exploring and creating camera motion for 3D scenes in augmented reality. Through AR, we leverage the spatial reference of the physical space and therefore enable consideration of the space's constraints. We designed CamARa as a web-based tool available on compatible mobile devices. It allows designers to intuitively create camera motion by seeing through both virtual and physical space without disorientation. CamARa also visualizes camera paths in an augmented environment, supporting deep exploration of shots and collaboration. Ultimately, CamARa offers an immersive experience that bridges the gap between physical and virtual worlds. We demonstrate CamARa to illustrate how spatial reference expands

¹<https://www.blender.org/>

the possibilities of a virtual camera system as a starting point for AR integration approaches.

Acknowledgments

This research was partially supported by the MSIT (Ministry of Science and ICT), Korea, under the ITRC (Information Technology Research Center) support program(IITP-2024-RS-2024-00436398) supervised by the IITP (Institute for Information & communication Technology Planning & evaluation). This work utilizes the 3D assets ‘Character Fight’² by LowPolyBoy and ‘House on Lake’³ by Bogdan Strielecki for demo purposes. Each model is provided under a CC Attribution-NonCommercial license, respectively.

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²<https://skfb.ly/oUuNx>

³<https://skfb.ly/otNyR>